

CV2060: Transportation Engineering

Term Project: Traffic Signal Timing Design

Group 2 | May 2026

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1. Introduction

This report presents a signal timing design for a four-leg signalized intersection assigned to Group 2. The intersection has four approaches (Northbound, Southbound, Eastbound, Westbound), each with two through lanes, one left-turn lane, and one right-turn lane.

The objective is to design 2-phase and 4-phase signal timing plans using Webster's method, evaluate their performance using average control delay and degree of saturation, and recommend the better phasing scheme.

2. Input Data

2.1 Traffic Volumes

Table 1 shows the movement volumes for Group 2.

Approach	Left Turn(veh/h)	Through(veh/h)	Right Turn(veh/h)	Sat. Flow LT(veh/h)	Sat. Flow T(veh/h)
Northbound (NB)	303	304	170	1800	2200
Southbound (SB)	269	403	154	1800	2200
Eastbound (EB)	39	608	106	1800	2200
Westbound (WB)	20	386	82	1800	2200

Table 1: Group 2 traffic volumes and saturation flow rates. RT saturation flow = 1500 veh/h/lane.

2.2 Design Assumptions

Parameter	Value
Lost time per phase (l)	3 sec
All-red interval per phase	1 sec
Pedestrian crossing time	12 sec (minimum green)
Speed limit (u_0)	45 km/h (12.5 m/s)
Reaction time (δ)	1 sec
Deceleration rate (a)	3 m/s ²
Intersection width (W)	15 m (standard geometry)
Average vehicle length (L)	4 m
Approach grade	Level (0%)

Table 2: Design parameters and assumptions.

3. Intersection Layout

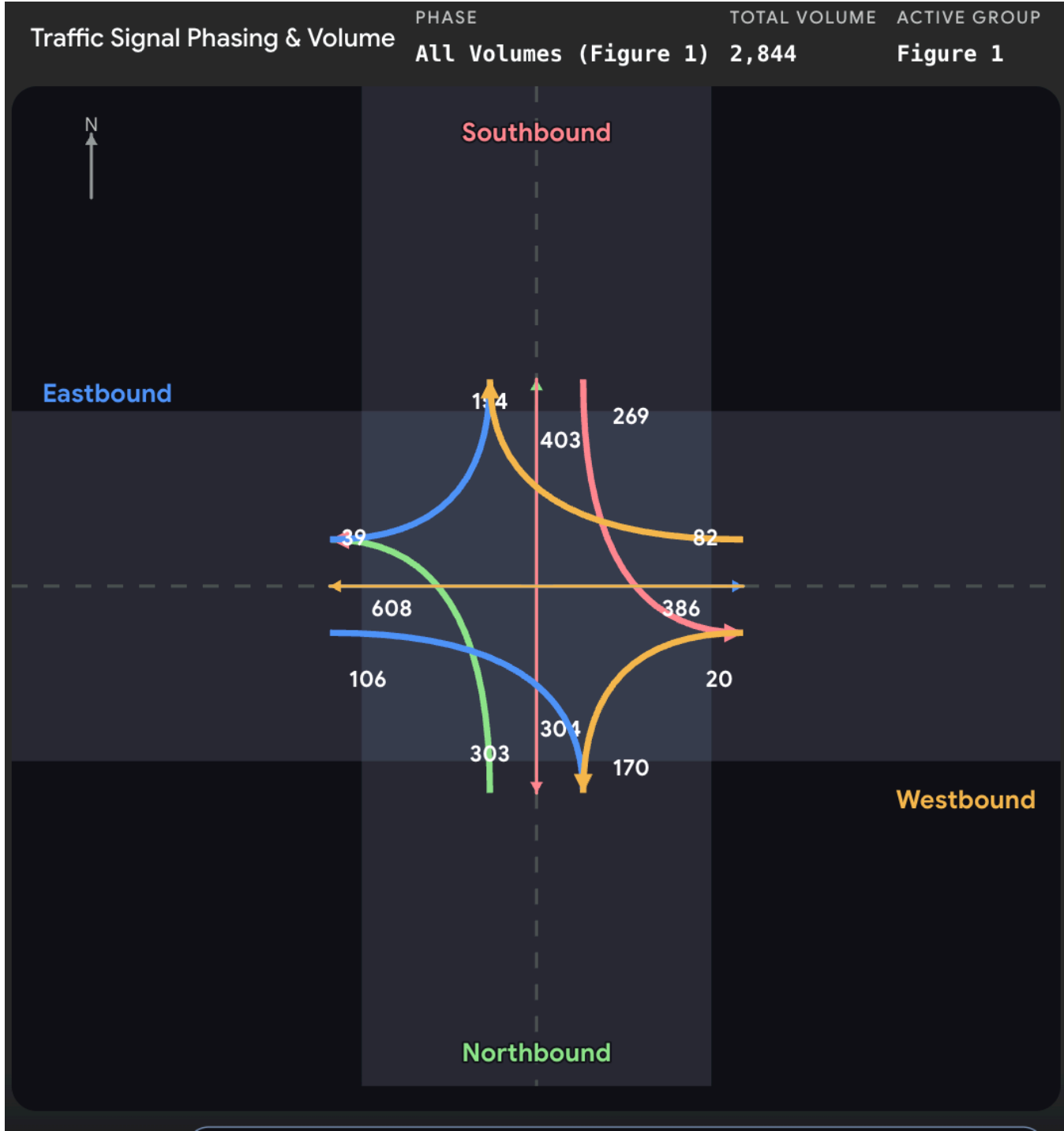


Figure 1: Four-leg intersection with directional movement volumes (veh/h) for Group 2. NB = green, SB = red, EB = blue, WB = orange.

4. Yellow Time Calculation

The minimum yellow interval is calculated from the dilemma zone formula to ensure vehicles can either stop safely or clear the intersection:

$$\tau_{\text{men}} = \delta + (W + L)/u_0 + u_0 / (2a)$$

$$\tau = 1 + (15 + 4)/12.5 + 12.5/(2 \times 3) = 1 + 1.52 + 2.08 = 4.60 \text{ sec} \rightarrow \text{use 5 sec}$$

A yellow interval of 5 sec is applied to all phases in both the 2-phase and 4-phase designs.

5. Two-Phase Signal Design

5.1 Phase Structure

Phase 1 serves N-S traffic (NB + SB simultaneously with permissive left turns). Phase 2 serves E-W traffic (EB + WB).

5.2 Critical Lane Volumes (Y values)

$Y_i = \max(q/s)$ for all lane groups in phase i. The critical movement in each direction sets the Y value.

Movement	Volume q(veh/h)	Sat. Flow s(veh/h)	q/s	Critical?	Phase Y	Phase
NB Left Turn	303	1800	0.168	No		1
NB Through	304	2200	0.138	No		1
NB Right Turn	170	1500	0.113	No		1
SB Left Turn	269	1800	0.149	No	$Y_1 = 0.183$	1
SB Through	403	2200	0.183	YES (max SB)		1
SB Right Turn	154	1500	0.103	No		1
EB Left Turn	39	1800	0.022	No		2
EB Through	608	2200	0.276	YES (max EB)		2
EB Right Turn	106	1500	0.071	No	$Y_2 = 0.276$	2
WB Left Turn	20	1800	0.011	No		2
WB Through	386	2200	0.175	No		2
WB Right Turn	82	1500	0.055	No		2

Table 3: Critical lane volume calculation for 2-phase design.

5.3 Optimal Cycle Length

$$\Sigma Y = Y_1 + Y_2 = 0.183 + 0.276 = 0.459$$

$$\text{Total lost time: } L = 2 \text{ phases} \times (I + AR) = 2 \times (3 + 1) = 8 \text{ sec}$$

$$\text{Webster's optimal cycle: } C_0 = (1.5L + 5) / (1 - \Sigma Y) = (1.5 \times 8 + 5) / (1 - 0.459) = 17 / 0.541 = 31.23 \text{ sec} \rightarrow \text{round to 35 sec}$$

$$\text{Total effective green: } G_{te} = C - L = 35 - 8 = 27 \text{ sec}$$

Effective green per phase:

$$G_{e_1} = (0.183/0.459) \times 27 = 10.7 \text{ sec} \rightarrow \text{Actual } G_1 = 10.7 + 3 - 5 = 8.7 \text{ sec} (< 12 \text{ sec})$$

$$G_{e_2} = (0.276/0.459) \times 27 = 16.3 \text{ sec} \rightarrow \text{Actual } G_2 = 16.3 + 3 - 5 = 14.3 \text{ sec} (< 12 \text{ sec})$$

Both phases fall below the 12 sec pedestrian minimum, so we increase the cycle to $C = 50$ sec and redistribute:

$$G_{e_1} = (0.183/0.459) \times 42 = 16.7 \text{ sec} \rightarrow G_{21} = 16.7 + 3 - 5 = 14.7 \approx 15 \text{ sec} \checkmark$$

$$G_{e_2} = (0.276/0.459) \times 42 = 25.3 \text{ sec} \rightarrow G_{22} = 25.3 + 3 - 5 = 23.3 \approx 23 \text{ sec} \checkmark$$

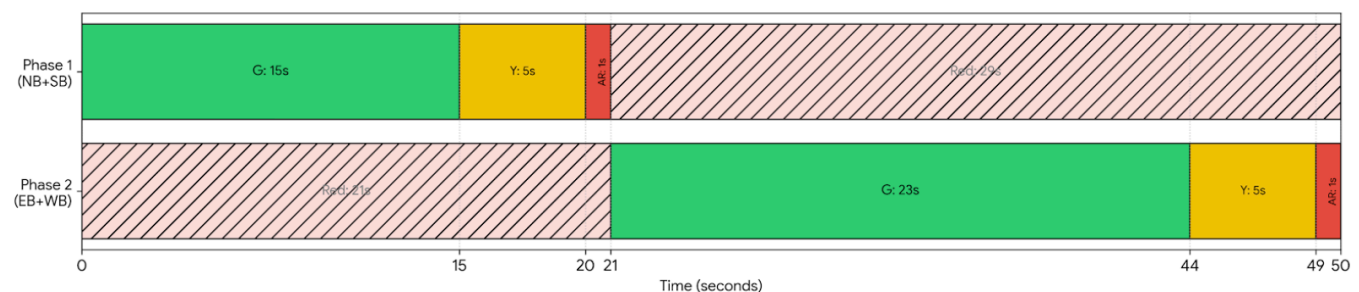
Cycle check: $15 + 5 + 1 + 23 + 5 + 1 = 50 \text{ sec} \checkmark$

5.4 Final 2-Phase Timing

Phase	Movements	Green (sec)	Yellow (sec)	All-Red (sec)	Phase Total (sec)
Phase 1	NB + SB	15	5	1	21
Phase 2	EB + WB	23	5	1	29
Total		38	10	2	50

Table 4: 2-phase signal timing plan (Cycle = 50 sec).

2-Phase Signal Timing Diagram (Cycle Length = 50 sec)



6. Four-Phase Signal Design

6.1 Phase Structure

Each approach is given an exclusive phase: Phase 1 = NB only, Phase 2 = SB only, Phase 3 = EB only, Phase 4 = WB only. This protects all left turns.

6.2 Critical Lane Volumes (Y values)

Phase	Controlling Movement	Volume (veh/h)	Sat. Flow (veh/h)	Y_i
Phase 1 (NB)	NB Left Turn	303	1800	0.168
Phase 2 (SB)	SB Through	403	2200	0.183
Phase 3 (EB)	EB Through	608	2200	0.276
Phase 4 (WB)	WB Through	386	2200	0.175

Table 5: Critical Y values for 4-phase design. $\Sigma Y = 0.168 + 0.183 + 0.276 + 0.175 = 0.802$

6.3 Optimal Cycle Length

$$\Sigma Y = 0.802$$

$$\text{Total lost time: } L = 4 \times (3 + 1) = 16 \text{ sec}$$

Webster's optimal cycle: $C_0 = (1.5 \times 16 + 5) / (1 - 0.802) = 29 / 0.198 = 146.46 \text{ sec} \rightarrow \text{round to } 150 \text{ sec}$

$$G_{te} = 150 - 16 = 134 \text{ sec}$$

Proportional actual green times without pedestrian constraint:

$$\text{NB: } G_2 = (0.168/0.802) \times 134 + 3 - 5 = 26 \text{ sec} > 12 \text{ sec} \rightarrow \text{pedestrian constraint satisfies}$$

$$\text{SB: } G_2 = (0.183/0.802) \times 134 + 3 - 5 = 29 \text{ sec} > 12 \text{ sec} \rightarrow \text{pedestrian constraint satisfies}$$

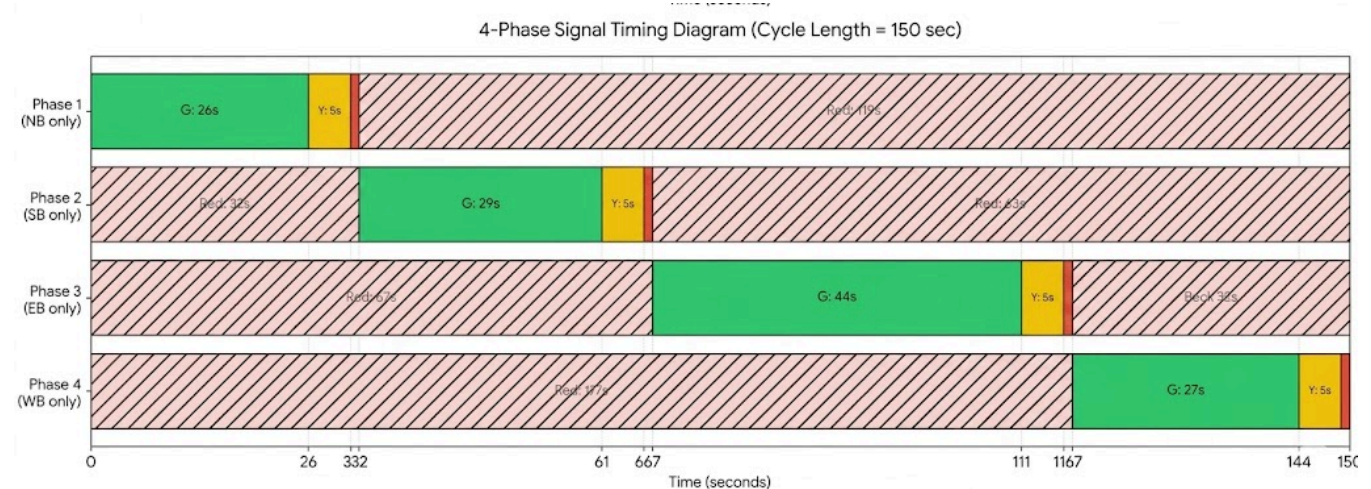
$$\text{EB: } G_2 = (0.276/0.802) \times 134 + 3 - 5 = 44 \text{ sec} > 12 \text{ sec} \rightarrow \text{pedestrian constraint satisfies}$$

$$\text{WB: } G_2 = (0.175/0.802) \times 134 + 3 - 5 = 27 \text{ sec} > 12 \text{ sec} \rightarrow \text{pedestrian constraint satisfies}$$

6.4 Final 4-Phase Timing

Phase	Movements	Green (sec)	Yellow (sec)	All-Red (sec)	Phase Total (sec)
Phase 1	NB only	26	5	1	32
Phase 2	SB only	29	5	1	35
Phase 3	EB only	44	5	1	50
Phase 4	WB only	27	5	1	33
Total		126	20	4	150

Table 6: 4-phase signal timing plan (Cycle = 150 sec)



7. Performance Evaluation

7.1 Degree of Saturation

Degree of saturation $x = (q/s) / (g_e/C)$, where g_e is effective green time. For 2-phase: $G_{e_1} = 17$ sec, $G_{e_2} = 25$ sec. For 4-phase: $G_{e_1} = 28$, $G_{e_2} = 31$, $G_{e_3} = 46$, $G_{e_4} = 29$ sec.

Movement	q (veh/h)	s (veh/h)	g_e/C (2-ph)	x (2-ph)	g_e/C (4-ph)	x (4-ph)
NB Left Turn	303	1800	0.34	0.49	0.186	0.91
NB Through	304	2200	0.34	0.41	0.186	0.74
NB Right Turn	170	1500	0.34	0.33	0.186	0.61
SB Left Turn	269	1800	0.34	0.44	0.206	0.72
SB Through	403	2200	0.34	0.54	0.206	0.89
SB Right Turn	154	1500	0.34	0.30	0.206	0.49
EB Left Turn	39	1800	0.5	0.06	0.306	0.07
EB Through	608	2200	0.5	0.55	0.306	0.90
EB Right Turn	106	1500	0.5	0.14	0.306	0.23
WB Left Turn	20	1800	0.5	0.02	0.193	0.057
WB Through	386	2200	0.5	0.35	0.193	0.91
WB Right Turn	82	1500	0.5	0.11	0.193	0.03

Table 7: Degree of saturation for all movements. All values < 1.0 (intersection not oversaturated). Values > 0.70 are highlighted.

7.2 Average Control Delay (Webster's Uniform Delay)

Webster's delay formula: $d = C(1-\lambda)^2 / [2(1-\lambda x)] + x^2 / [2q(1-x)]$

where $\lambda = g_e/C$ (effective green ratio), q = flow rate in veh/sec, x = degree of saturation.

Approach-level delay is volume-weighted average of individual movement delays.

Approach	Delay 2-Phase (sec/veh)	LOS	Delay 4-Phase (sec/veh)	LOS
Northbound	14.9	B	57.5	E
Southbound	15.4	B	48.8	D
Eastbound	9.9	A	43.4	D
Westbound	8.1	A	62.5	E
Intersection (avg)	12.5	B	52.1	D

Table 8: Average control delay and Level of Service (LOS) by approach and intersection. LOS thresholds: A ≤ 10, B 10–20, C 20–35, D 35–55, E 55–80 sec/veh.

7.3 Summary Comparison

Parameter	2-Phase	4-Phase
ΣY	0.459	0.802
Optimal Cycle (Webster's)	35 sec	150 sec
Pedestrian-adjusted Cycle	50 sec	150 sec
Yellow Time (per phase)	5 sec	5 sec
Avg. Intersection Delay	12.5 sec/veh	52.1 sec/veh
Intersection LOS	B	D
Max. Degree of Saturation	0.55 (EB-T)	0.91 (NB-LT, EB-T)

Table 9: Design summary comparison between 2-phase and 4-phase plans.

8. Conclusion

The 2-phase design is clearly more efficient for this intersection. It gives an average intersection delay of only 12.5 sec/veh (LOS B) compared to 52.1 sec/veh (LOS D) for the 4-phase design. The shorter cycle (50 sec vs 150 sec) means vehicles wait less on average, and all degrees of saturation remain well below 0.6. The intersection has plenty of capacity headroom.

That said, the 2-phase design uses permissive left turns for the heavy NB and SB left turn volumes (303 and 269 veh/h). If safety is a concern for these crossing movements, a protected-permissive phasing would be worth exploring as a middle ground. This could be a potential area for further study.

In conclusion, the 2-phase timing plan is recommended for this intersection based on its significantly lower delay and simpler operation.